FIRST NAME _____ SURNAME_____ CLASS_____

2020 CHINSAPO CLUSTER MOCK

2020 MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

PHYSICS

PAPER I

(100 marks)

Subject Number: M164/I

Time Allowed: 2 hours

Instructions

1. This paper contains **8 pages. Please check**.
2. Fill in your **examination number** and **school name** at the top of each page.
3. This paper contains two **sections A and B**. in Section A there are eight short answer questions while in section B there are five restricted essay questions.
4. Answer all the thirteen questions in the spaces provided.
5. Use of electronic calculators is allowed.
6. The **maximum number** of marks for each answer is indicated against each question.

Question Number	**Tick if answered**	**Do not write In these columns**	
1			
2			
3			
4			
5			
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7			
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10			
11			
12			
13			

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Section A (70 marks)

1. a. State any two branches of physics.

_____

_____ (2 marks)

b. Force is defined as mass multiplied by acceleration (F = ma). The SI unit of force is a Newton. What is 1N in base units?

_____ (2 marks)

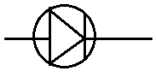
c. Describe the scientific approach employed in the study of physics.

_____

_____

_____ (3 marks)

2. a. What electrical components are represented by the symbols below?

i.  _____ (1 mark)

ii. _____ (1 mark)

b. In a circuit, a 4Ω resistor and an 8Ω resistor are connected in parallel and a 2Ω resistor is connected in series with them.

i. Draw a circuit diagram using the given information.

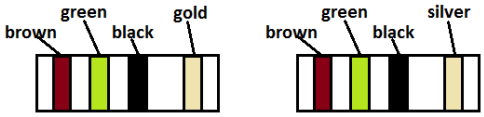
(2 marks)

ii. Calculate the total resistance in the circuit in 2b(i) above.

c. Figure 1 is a diagram showing colour codes and resistors marked A and B.

Colour	Black	Brown	Red	Orange	Yellow	Green
Code	0	1	2	3	4	5

Tolerance: Gold = 5% Silver = 10%



i. What is the resistance of resistor A?

_____ (2 marks)

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ii. What is the advantage of using resistor A in the circuit than resistor B?

_____

_____ (2 marks)

3. a. State two differences between a water wave and sound wave.

_____

_____

_____

_____ (4 marks)

b. With the aid of a diagram describe constructive interference.

_____

_____

_____(3 marks)

c. The diagram below shows a wave.

i. What type of wave is shown above? Give a reason for your answer.

_____

_____ (2 marks)

ii. Calculate wavelength of the wave above.

(2 marks)

iii. Calculate the frequency if the 50cm distance is covered in 5 seconds.

(3 marks)

d. i. Define ‘focal length of a convex lens

_____

_____(1 mark)

ii. An object is placed 10cm away from a lens of focal length 5cm.
use the lens formula to find the image distance.

(3 marks)

e. Which parts of the camera works similar to the parts of the eye below

i. Retina? _____ (1 mark)

ii. Iris ? _____ (1 mark)

4. a. What is the difference between 'heat capacity' and 'specific heat capacity'?

_____

_____

_____

_____ (2 marks)

b. How much energy is needed to increase the temperature of 500g of lead from 20^oC to 45^oC? (specific heat capacity of lead is $128J/Kg^oC$)

(4 marks)

c. i. Give any two applications of specific heat capacity in everyday life.

_____

_____ (2 marks)

ii. State any three ways in which heat is transferred.

_____

_____

_____ (3 marks)

5. a. State two factors that affect pressure exerted by solids.

_____

_____(2 marks)

b. Why are dam walls made broader at the base than at the top?

_____

_____

_____ (2 marks)

c. A pressure of 50000Pa is exerted by a column of water on the base of a container. Calculate the height of the water column. (density of water = 1000Kg/m^3, g = ms^{-2})

(2 marks)

d. Why is a hydraulic press called a force multiplier?

_____

_____

_____(2 marks)

6. a.State any two forces which act against an object when it is falling in a liquid.

_____

_____(2 marks)

b. State Newton's second law of motion.

_____

_____

_____ (1 mark)

c. A car of mass 2000Kg accelerates uniformly from rest to a velocity of 10m/s in 5 seconds. Calculate;

i. acceleration of the car

(2 marks)

ii. distance covered by the car

(2 marks)

iii. force required to produce the acceleration.

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(2 marks)

7. a. Define uniform circular motion

_____

_____(1 mark)

b. How does the centripetal force change if an object

i. has more mass?

_____ (1 mark)

ii. Travels at a slower speed?

_____ (1 mark)

iii. Travels round a tight curve?

_____ (1 mark)

8. a. What is a 'semi – conductor'?

_____

_____ (1 mark)

b. What is doping in relation to semi – conductors?

_____

_____ (1 mark)

c. State two functions of transistors

_____

_____ (2 marks)

Section B (30 marks)

9. a. With the aid of a diagram, describe how a step up transformer works.

_____

_____

_____

_____

_____(4 marks)

b. State two ways in which energy loses in a transformer are minimized.

_____

_____

_____ (2 marks)

10. a. What is nuclear fission?

_____

_____ (1 mark)

b. Explain how gamma rays are emitted.

_____

_____

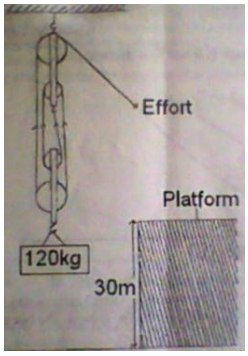
_____

_____ (4 marks)

c. Give one way of detecting radioactive particles

_____ (1 mark)

11. Figure 2 is a diagram of a pulley being used to lift a mass of 120Kg vertically through a height of 30m.



i. Calculate the work done in lifting the mass to a height of 30m.

(3 marks)

ii. How much energy is supplied by the effort in raising the mass through the height of 30m?

_____ (1 mark)

iii. If it takes 0.5 minutes to complete the work using the pulley system, work out the power of this machine.

(3 marks)

12. a. Define ‘efficiency’ in relation to machines

_____

_____ (1 mark)

b. Give three reasons why in practice the machine in figure 1 is not 100% efficient.

_____

_____ (3 marks)

13. in a thermos flask ,explain the function of

a. Silvered glass

_____

_____

_____

_____(2 marks)

b. Vacuum

_____

_____

_____

_____(2marks)

END OF QUESTION PAPER